INCH-POUND
MIL-DTL-24643/13E
1 October 2009
SUPERSEDING
MIL-DTL-24643/13D
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +90 °C, 300 VOLTS, TYPE LSTTRS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Non-Watertight

First	-	Copper conductor, coated or uncoated, 20 AWG, Class B stranding in accordance with ASTM B8.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation. Nominal wall thickness of 0.020 inch; wall thickness may vary from nominal as necessary, in order to meet the specified electrical requirements.
Fourth	-	Two conductors, one black and one white (or natural), shall be cabled together to form a pair. The length of lay shall be determined to permit a completed cable to meet the twisting and bending endurance tests.
Fifth	-	An optional binder.
Sixth	-	Braided shield of 34 or 36 AWG, coated or uncoated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.
Seventh	-	Shield insulation of two polyester tapes with the outer tape sealed. An alternate shield insulation may consist of one polyester tape plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by Method 2.
Eighth	-	The specified number of pairs (see table I) shall be cabled together. The length of lay(s) shall be required for the completed cable to meet the twisting and bending endurance tests. Cabling sequence to be consecutive, starting with No. 1, from the center outward. A single pair shall not be used as center for the cable core. Outer layer shall be left hand. Inner layer (s) direction of lay shall be at the manufacturer's option. Fillers may be used as necessary to form a firm, well rounded assembly.
Ninth	-	An optional binder.
Tenth	-	Cross-linked polyolefin jacket (see table I for thickness).

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TABLE I. Details.

Military part no. M24643/13	Type and size	Number of pairs	Jacket wall thickness min. avg. (inch)	Overall diameter (inch)		Roller diameter (inch) (max.)
				min. (inch)	max. (inch)	
-01UN	LSTTRS-2	2	0.050	0.629	0.680	2.0
-02UN	LSTTRS-4	4	0.050	0.685	0.740	2.2
-03UN	LSTTRS-6	6	0.050	0.814	0.880	3.0
-04UN	LSTTRS-8	8	0.050	0.916	0.990	3.5
-05UN	LSTTRS-10	10	0.050	0.999	1.080	4.0
-06UN	LSTTRS-12	12	0.070	1.017	1.100	4.0
-07UN	LSTTRS-16	16	0.070	1.100	1.190	4.5

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	11.95
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2000
Conductor to shield	1000
Component shield to component shield	200
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	500
Conductor to shield	200
Component shield to component shield	100
Conductor continuity	No failure
Shield continuity	No failure

Group A:

Visual and dimensional	No failure
Capacitance	
Mutual capacitance – at 1 megahertz (MHz) (pF/ft, max.)	28
Capacitance unbalance (percent, max.)	10
Characteristic impedance – at 1 MHz (ohms)	
Minimum	75

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Maximum	90
Group B:	
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Twisting and bending endurance (see table I for roller size)	
At 75 °C (cycles, min.)	6000
At minus 30±2°C (cycles, min.)	2000
Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) air oven	
Insulation (extruded)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Shield - conformance to material, construction and coverage	No failure
Group D:	
Flame propagation (cable)	No failure

Qualification Inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (125±5°C)	No failure

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Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSTTRS all sizes	1500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-017)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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